

Arije Trabelsi - Business Intelligence Student

Seeking a 3 to 4-month end-of-studies internship (PFE), starting January 2027

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Profile

Third-year Business Intelligence student at ESEN, looking for a 4-month end-of-studies internship (PFE) starting January 2027. I want to contribute to a real-world decision-support project within a data team, from data preparation to actionable insights.

Education

ESEN (Higher School of Digital Economy), Tunisia 2024 - Present
Bachelor's Degree in Business Intelligence

Medjez El Bab High School 2020 - 2024
Baccalaureate in Experimental Sciences

Professional Experience

Summer Intern – TeachCodeX July - August 2026
Co-developed Nexus 4.0, a full-stack decision-support platform (role-based dashboards, AI chatbot, reporting engine)

- Built a role-aware AI chatbot (admin, employee, client) that answers natural-language questions on KPIs, tickets and reports using RAG (BM25 retrieval + LLM via the Groq API).
- Designed a smart ticket-assignment algorithm: weighted scoring (60% workload, 40% skills), prioritization by severity and age, with explainable recommendations.
- Developed the reporting engine with FastAPI, SQLAlchemy and PostgreSQL, tracking 5 ISO 9001-aligned KPIs with RAG status and history.
- Implemented synchronous and asynchronous PDF and Excel report generation (xhtml2pdf, openpyxl), with files stored on Cloudinary.
- Set up risk alerts driven by a predictive score, with in-app and email notifications (SMTP).
- Secured API routes with JWT authentication and role-based access control.

Technical Skills

Languages: Python (Pandas, NumPy, Matplotlib, Seaborn), Java, JavaScript, PHP, C, HTML, CSS
Databases: PostgreSQL, Oracle SQL, PL/SQL, MySQL, SQLite
Web Frameworks: FastAPI, SQLAlchemy, React.js, Node.js, Chart.js, Odoo
BI & Data Visualization: Power BI (DAX), Tableau, Talend, R (RStudio), Excel (Pivot Tables)
APIs & AI: RESTful APIs, JWT, RAG, BM25, Groq API, OpenAI API, Claude API (LLM integration, NLP)
Algorithms & Decision Support: Weighted scoring, recommendation systems, multi-criteria prioritization
Modeling & Tools: UML, BPMN, BonitaSoft, Scrum, Git, Postman, Cloudinary, MQTT, Node-RED
Soft Skills: Problem solving, UI/UX design, Analytical thinking

Projects

Sales Analytics BI Solution | *Talend*, *RStudio*, *ML*, *PostgreSQL*, *Power BI* [Project Report]
End-to-end decision-support solution built on a retail sales dataset: data warehouse, ETL pipeline, dashboards and machine learning.

iTunes vs Spotify Comparative Analysis | *Python*, *Pandas*, *Seaborn*, *Power BI*, *DAX* [GitHub]
End-to-end pipeline on about 20K tracks: data cleaning and EDA in Python, then a 3-page Power BI dashboard with custom DAX measures and dynamic KPIs.

Employee Attrition Analysis (IBM HR) | *Python*, *Pandas*, *Seaborn*, *SciPy* [GitHub]
Exploratory analysis of employee turnover drivers: correlation analysis, hypothesis testing (t-test, chi-square) and outlier detection (IQR).

FoodLoop | *JavaScript*, *PHP*, *HTML*, *CSS*, *Chart.js*, *PL/SQL*, *Oracle* [GitHub] [Live Demo]
Web platform connecting users, charities and food retailers to manage surplus food and reduce waste.

InternConnect | *JavaScript*, *HTML*, *CSS* [GitHub]
Front-end prototype where students apply to internship offers and companies manage their applications.

Languages

Arabic: Native **French**: Advanced **English**: Fluent **Spanish**: Beginner